Prompt Document: Clinic Appointment Website with AI Chatbot

Candidate: Harini
Category: Website Development (Healthcare)
AI tools used: Claude (planning, coding, debugging), Groq API (chatbot answers)
Tech stack: FastAPI (Python) backend, TF-IDF retrieval (scikit-learn), SQLite database, web frontend

1. Understanding the assessment

Prompt 1: "explain this assessment" (I pasted the full assessment email.)

Why: I wanted to be sure I understood the three required submissions (video, prompt document, project link) and how my work would be judged.

Result: I got a simple explanation of the task, the five categories and the judging criteria.

2. Choosing the project

Prompt 2: "I choose web development"

Result: I received three healthcare and technology project ideas. I chose the clinic appointment booking website because it solves a real problem and is easy to demonstrate.

3. Planning and building

Prompt 3 (first, broad): "Help me build a clinic appointment booking website."

Result: A basic plan and project structure. It was too general, so I made the next prompts more specific.

Prompt 4 (improved, features): "Build the clinic website with a doctor list, a page to book an appointment where I choose a doctor, a day the doctor works and a free time slot, then enter the patient's name and phone number, and get a confirmation with a reference number starting with CLN."

Result: The booking flow and database tables for doctors and appointments.

Prompt 5 (improved, AI chatbot): "Add a chatbot to the website that answers patient questions about the clinic, doctors and booking. Use a small knowledge base file and the Groq API for answers, with a fallback that works without the API."

Result: The chatbot pipeline: safety check, then retrieval from clinic_info.md, then answer generation.

Prompt 6 (improved, safety): "The chatbot must not give medical diagnoses. For symptoms like fever, it should give general guidance and a clear note that it is not medical advice, and suggest the right type of doctor."

Result: Safety rules for symptom questions such as "I have fever", "my child has a fever" and "pregnant lady".

Prompt 7 (follow-up, setup): "Give me step-by-step instructions to run the backend and frontend on Windows."

Result: Setup steps, including running the backend with `uvicorn main:app --reload --port 8000`.

4. Debugging the chatbot

Problem: When I asked "an appointment", the chatbot showed vaccination information instead of the booking steps.

Prompt 8 (reporting the error): "When I ask 'an appointment' the chatbot shows vaccination information, not how to book. I replaced rag.py but I still get the same answer."

Result: Claude suggested that my backend might not be running the new file, and asked me to check the running folder and file names.

Prompt 9 (follow-up, checking files): I shared the folder path of the running backend and the results of the `findstr` commands.

Result: This showed I needed to share my real code instead of describing it.

Prompt 10 (improved, sharing real code): "i will give chatbot.py code,rag.py code you can see this ..then i will fix this" (I pasted the full rag.py and chatbot.py.)

Result: Claude tested my real files and found the cause: the section "How to book an appointment online" was missing from clinic_info.md, so the retriever chose the vaccination section, which also contains the words "book" and "appointment". The fixes were:

- Added a built-in booking section in rag.py that is used if the section is missing from the knowledge base.
- Made simple booking questions such as "appointment" and "how do I book" always show the booking steps and the "Book an appointment" button.
- Removed one synonym line that made every appointment question show booking steps, so questions like "how long is an appointment" and "how can I cancel an appointment" now get their own answers.
- Added the keywords "vaccine" and "vaccination" so vaccination questions find the right section.

Prompt 11 (verification): After replacing rag.py and restarting the backend, I tested "an appointment" and "how do I book an appointment".

Result: Both questions now return the correct booking steps. The fever, pregnancy and doctor list answers stayed the same.

5. Final steps

Prompt 12: "I completed the full project"

Result: I got a checklist of what to submit.

6. What I learned about prompting

1. Starting with a broad prompt and then adding exact features, rules and constraints gave much better results.
2. Pasting my real code and exact error messages worked better than describing the problem.
3. Testing after every fix showed which change actually solved the problem.

4 Asking for safety rules early (no diagnosis, clear medical note) made the chatbot more responsible.

7. Final result

A working clinic website where patients can view doctors, book an appointment and receive a CLN reference number, plus an AI chatbot that answers clinic questions correctly and handles symptom questions safely.